**A CASE-STUDY BASED REPORT**

**ON**

AI-Driven Stock Return Prediction Using Ridge and Lasso Regression with Feature Scaling

***submitted in partial fulfilment of the requirements for the Course***

# MACHINE LEARNING

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**by**


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# INDEX

**Abstract**

**Introduction** .....

Background .....
Motivation for the Project .....
Overview of the Work Carried Out .....
Scope and Responsible-Use Statement .....

**Problem Statement** .....

Case-Study Scenario .....
Central Case-Study Question .....
Formal Problem Definition .....
Objectives that Guide the Study .....
Assumptions and Limitations .....

**Dataset and Exploratory Data Analysis**

About the Dataset .....
Purpose of EDA .....
Data Preparation Steps .....
EDA Exploration - Implementation .....

**Algorithm Selection and Implementation** .....

**Conclusion**

**Signature of the Team members**

# ABSTRACT

The stock market produces large volumes of financial data that can be analysed using machine learning techniques to identify trends and estimate future stock returns. This case study, titled "AI-Driven Stock Return Prediction Using Ridge and Lasso Regression with Feature Scaling," focuses on building an interpretable regression-based workflow for estimating stock returns from historical market data and engineered financial features.

The workflow described for the project includes data collection, preprocessing, exploratory data analysis (EDA), feature engineering, and feature scaling. Ridge Regression and Lasso Regression are used to study how regularisation can reduce overfitting and improve generalisation. The models are evaluated through a train-test split, k-fold cross-validation, and the regression metrics Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and $R^2$ score. Hyperparameter tuning is included to select a suitable regularisation strength.

For usability, the project describes a Streamlit dashboard for visualising historical stock trends, comparing Ridge and Lasso predictions, and generating return forecasts. Stock predictions are estimates and should not be treated as financial advice.

# 1. INTRODUCTION

## 1.1 Background

Financial markets generate time-dependent data such as opening price, highest price, lowest price, closing price, and trading volume. Analysing these observations can help identify patterns and quantify changes in market value. Machine learning provides methods for learning relationships between historical inputs and a target variable. In financial modelling, predicting stock return—the percentage change in value over a period—is different from predicting an exact future price and can provide a normalised measure for comparing changes over time.

Linear regression is a useful baseline because its predictions can be interpreted through feature coefficients. However, when input variables are numerous or correlated, an ordinary linear model may fit noise in the training data. Ridge and Lasso regression address this issue by adding regularisation penalties to the model objective. Feature scaling is important because regularisation is sensitive to the relative magnitudes of input features.

## 1.2 Motivation for the Project

The motivation is to examine whether regularised linear models can provide useful, interpretable estimates of stock returns from historical market information. The project also compares two different penalties: Ridge distributes shrinkage across coefficients, while Lasso can shrink some coefficients to zero. Comparing both approaches provides a practical case study of regularisation, feature scaling, model validation, and performance measurement in machine learning.

## 1.3 Overview of the Work Carried Out

The project workflow is organised into data collection, preprocessing, EDA, feature engineering, feature scaling, model training, evaluation, hyperparameter tuning, and dashboard integration. Ridge Regression and Lasso Regression are the two principal algorithms. The models are evaluated with a train-test split, k-fold cross-validation, RMSE, MAE, and $R^2$ score. A Streamlit interface is described as the user-facing component for trend visualisation and prediction comparison.

## 1.4 Scope and Responsible-Use Statement

The scope is limited to supervised regression for estimating stock returns from historical market data and to comparing Ridge and Lasso under a consistent evaluation workflow. The dashboard is intended for learning, visualisation, and experimentation. Market conditions can change suddenly, historical relationships may not continue, and predictions can be inaccurate. The system does not guarantee returns and must not be used as the sole basis for investment decisions. Users should verify data quality, understand model limitations, and seek qualified financial advice when needed.

# 2. PROBLEM STATEMENT

## 2.1 Case-Study Scenario

Consider a user analysing historical stock-market observations who wants a data-driven estimate of a future stock return rather than only a chart of past prices. The user needs a repeatable process to prepare the data, construct informative features, train a model, and compare forecasts against observed outcomes. The case study examines how Ridge and Lasso regression can be applied to this task and how their performance can be evaluated systematically.

## 2.2 Central Case-Study Question

How effectively can Ridge Regression and Lasso Regression estimate stock returns from historical market data when features are appropriately prepared and scaled, and how can validation metrics be used to compare their generalisation performance?

## 2.3 Formal Problem Definition

Let X represent a matrix of input features derived from historical market observations, and let y represent the target stock return for the prediction horizon. The objective is to learn a function f(X) that estimates y for observations not used during training. The training process should minimise prediction error while controlling model complexity through regularisation. A chronological separation of training and testing data is appropriate for time-dependent market observations because future observations should not influence the fitting of a model evaluated on the past.

For a linear model, the prediction can be represented as $\hat{y} = \beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p$. Ridge adds an L2 penalty on coefficient magnitudes, whereas Lasso adds an L1 penalty. The regularisation strength controls the trade-off between fitting the training data and keeping coefficients constrained.

## 2.4 Objectives that Guide the Study

- Prepare historical stock-market data for supervised regression.
- Explore data patterns and relationships through exploratory data analysis.
- Create relevant financial features from available market observations.
- Apply feature scaling so that the regularisation penalties operate on comparable feature magnitudes.
- Train and compare Ridge Regression and Lasso Regression.
- Use train-test evaluation, k-fold cross-validation, and RMSE, MAE, and $R^2$ score to assess performance.
- Tune regularisation strength and present model outputs through a Streamlit dashboard.

## 2.5 Assumptions and Limitations

- Historical data is assumed to be sufficiently accurate and representative for the experiment.
- Features available at prediction time must not contain information from the future target period.
- Financial time series are non-stationary; relationships learned from earlier data may weaken or change.
- Regression models estimate relationships in the input features and cannot account for every market event.

# 3. DATASET AND EXPLORATORY DATA ANALYSIS

## 3.1 About the Dataset

The data basis is historical stock-market data and engineered financial features. Typical market records can include Open, High, Low, Close, and Volume values, with a date or timestamp that establishes the order of observations. Derived features may represent percentage returns, price ranges, moving averages, volatility, momentum, or volume behaviour.

## 3.2 Purpose of EDA

Exploratory Data Analysis is used to understand the dataset before modelling. It can reveal missing values, duplicate observations, unexpected values, changes in price and volume, outliers, and relationships among features. Visual inspection also helps identify whether features have very different scales and whether strong correlations may affect coefficient interpretation.

## 3.3 Data Preparation Steps

The workflow includes data collection, preprocessing, EDA, feature engineering, and feature scaling. The preparation procedure is as follows:

- Inspect column names, data types, chronological ordering, and missing-value counts.
- Remove or handle invalid and missing records using a consistent rule.
- Construct financial features from historical observations and define the return target and prediction horizon.
- Separate input features from the target variable and avoid including future information in the inputs.
- Split observations into training and testing sets; for time-series forecasting, preserve chronological order.
- Fit the scaler on training data only, then use the fitted scaler to transform validation and test data.

## 3.4 EDA Exploration – Implementation

A suitable EDA implementation examines summary statistics for numerical columns, plots historical closing prices and trading volume, checks missing values, and studies feature correlations. Distribution plots can show skewness and outliers, while a correlation heatmap can indicate redundant or strongly related inputs. These checks inform feature preparation but do not, by themselves, establish predictive performance.

# 4. ALGORITHM SELECTION AND IMPLEMENTATION

## 4.1 Why Ridge and Lasso Regression?

Ridge and Lasso are selected because both are linear regression techniques that add regularisation to reduce excessive coefficient magnitudes. They provide a useful comparison while keeping the model relatively interpretable. Ridge is often useful when several features contribute related information; Lasso can also perform feature selection by setting some coefficients exactly to zero.

## 4.2 Ridge Regression

Ridge Regression minimises the residual sum of squares together with an L2 penalty on the coefficients. Its objective can be written as: $\Sigma(y_i - \hat{y}_i)^2 + \alpha\Sigma\beta_j^2$. The parameter $\alpha$ controls the penalty strength. A larger $\alpha$ generally shrinks coefficients more strongly. Ridge usually retains all input features, although their coefficient magnitudes may become small.

## 4.3 Lasso Regression

Lasso Regression minimises the residual sum of squares together with an L1 penalty: $\Sigma(y_i - \hat{y}_i)^2 + \alpha\Sigma|\beta_j|$. The parameter $\alpha$ controls the strength of regularisation. Because the L1 penalty can drive coefficients to exactly zero, Lasso can produce a simpler model with fewer active features. Its selection behaviour can vary when features are strongly correlated.

## 4.4 Importance of Feature Scaling

Market features may use different units and magnitudes. For example, price values and trading volume may have much larger numerical ranges than percentage returns or indicator values. Standardisation commonly transforms each feature using $z = (x - \mu) / \sigma$, where $\mu$ and $\sigma$ are estimated from the training data. Scaling helps ensure that regularisation is not dominated merely by the units in which a feature is measured. The scaler must be fitted on training data only to prevent data leakage.

## 4.5 Training and Validation Workflow

1. Collect and order the historical observations.
2. Create the target return and the input features defined by the implementation.
3. Inspect and prepare the dataset, then separate the target from the feature matrix.
4. Split data into training and test portions; preserve time order where appropriate.
5. Fit feature scaling on the training portion and transform the remaining data with the same scaler.
6. Train Ridge and Lasso models using the training data.
7. Tune the regularisation parameter using cross-validation and a suitable search procedure.
8. Evaluate the selected models on held-out observations and compare the metrics.
9. Save reusable preprocessing/model artefacts if supported by the implementation, and connect predictions to the dashboard.

## 4.6 Evaluation Metrics

Mean Absolute Error (MAE) is the average absolute difference between actual and predicted returns. It is easy to interpret in the units of the target. Root Mean Squared Error (RMSE) is the square root of the average squared error and penalises larger errors more heavily. $R^2$ score compares model error with the variation in the target; a value closer to 1 generally indicates a better fit on the evaluated data, while a negative value can occur when predictions perform worse than a constant-mean baseline.

### 4.7 Dashboard

The project includes a React-based dashboard that presents historical stock trends, compares Ridge and Lasso predictions, and supports return forecasts through an interactive interface. A modular design separates data preprocessing, model training, evaluation, and prediction. This separation makes the workflow easier to maintain and allows the user interface to call the prediction pipeline without mixing every operation in one script.

### 4.8 Implementation and Reproducibility Notes

The software requirements include Windows 10/11 or Ubuntu 22.04, Python 3.11 or above, Visual Studio Code or Jupyter Notebook, Git/GitHub, NumPy, Pandas, Matplotlib, Seaborn, Plotly, scikit-learn, yfinance, Joblib, Streamlit, and Google Chrome. The final implementation should record the exact package versions and data retrieval date where possible, because financial data sources and historical records may change over time.

# 5. CONCLUSION

This case study presents a structured machine learning approach to stock return prediction using Ridge Regression and Lasso Regression with feature scaling. It connects the main stages of a practical regression workflow: historical data preparation, EDA, feature engineering, scaling, regularised model training, hyperparameter tuning, validation, and interactive presentation through Streamlit.

The two algorithms offer complementary forms of regularisation. Ridge shrinks coefficient magnitudes, while Lasso may also remove less useful features by setting coefficients to zero. A fair comparison requires consistent preprocessing, careful separation of training and test data, and suitable validation. MAE, RMSE, and $R^2$ provide different views of prediction quality and should be interpreted together.

Stock-return forecasts are uncertain estimates and not guarantees of future performance.

**Signature of the Team Members**

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